

GONZALO CRUZ GÓMEZ

gonza.c.gomez03@gmail.com | github.com/gonzalo-cruz | gonzalo-cruz.github.io/portfolio/

Data Science & Engineering undergraduate (3rd year, URJC) with a background in Telecommunications Engineering. Builds production-ready data systems: end-to-end ML pipelines, distributed computing, and deep learning. Founder of Asociación Datalab. Seeking an internship in data science or data engineering.

EDUCATION

- | | | |
|---|----------------------|-----------------------|
| B.S. Data Science & Engineering | URJC | <i>2024 – Present</i> |
| • Completed two academic years in one (2024–2025). Coursework: ML, Distributed Systems, Deep Learning. | | |
| B.S. Telecommunications Engineering | UC3M | <i>2019 – 2023</i> |
| • 120+ credits in Networking, Physics and Signal Processing. Transferred to specialize in Data Science. | | |
| Certificate of Advanced English (C2) | Cambridge Assessment | <i>2019</i> |

PROJECTS

- | | |
|---|--|
| TripAdvisor Restaurants Data Pipeline | <i>Python, Apache Airflow, Kafka, Scikit-learn, Pandas</i> |
| • Orchestrated a 5-task ETL pipeline with Apache Airflow, processing 1M+ rows in 50k-row chunks to stay memory-safe throughout — extract, clean, EDA, preprocessing, and load. | |
| • Fitted <code>IncrementalPCA</code> and <code>StandardScaler</code> via <code>partial_fit</code> on batches; applied OHE or label encoding per column based on cardinality. | |
| • Streamed the processed dataset to Kafka using a fault-tolerant producer (<code>acks=all</code> , sub-batch delivery) ready for downstream consumption. | |
| • Centralised all pipeline parameters (chunk size, thresholds, Kafka config) in a single <code>config.toml</code> ; all tasks read from it at runtime. | |
| Forest Fire Spread Prediction | <i>R, GLMs, GAMs, Statistical Modeling</i> |
| • Built GLM and GAM regression models to predict fire-burned area from meteorological covariates; GAMs outperformed GLMs on held-out data with lower residual variance. | |
| • Resolved severe data quality issues (high skew, missing values, near-zero-variance predictors) via log transformation, imputation, and feature selection. | |
| • Applied cross-validation to compare model fits; selected GAMs as the final model based on lower RMSE and better capture of non-linear covariate effects. | |
| Stroke Risk Classification | <i>Python, Scikit-learn, TensorFlow, ANN</i> |
| • Trained an MLP on an imbalanced medical dataset, applying SMOTE and optimizing the decision threshold to minimize false negatives given the clinical cost of missed strokes. | |
| • Selected AUC-ROC and F1 as evaluation metrics over accuracy; tuned architecture and hyperparameters accordingly. | |
| • Benchmarked the MLP against a logistic regression baseline; the non-linear model improved recall on the stroke class while maintaining acceptable precision. | |

EXPERIENCE

- | | |
|--|---------------------------|
| Founder & President — Asociación Datalab | <i>May 2025 – Present</i> |
| Established the university's first Data Science student association: defined the strategic roadmap, handled legal registration, and organized technical workshops. | |
| Private Tutor — Mathematics & Physics | <i>Sep 2021 – Present</i> |
| Mentored high-school students over 5 years, achieving a 100% pass rate and helping 90% of students improve by at least 3 grade points through personalized plans. | |

TECHNICAL SKILLS

- | | |
|----------------------|---|
| Programming: | Python (Advanced), R, SQL, C |
| Data Science: | Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Hadoop, Spark, Matplotlib |
| Engineering: | Apache Airflow, Kafka, Docker, Git/GitHub, Linux (Bash) |
| Languages: | Spanish (Native), English (C2), German (B1), Chinese (HSK3) |